

Reviewer 1

The submitted paper is interesting and clearly written. It is overall a good work. I only have two concerns.

1. The study area is not very large and seems environmentally homogeneous. My own experience indicates that the larger the study area, and the environmental heterogeneity, the most difficult is to train a good classification model. This issue is especially serious when topographical variables are added as predictors. Do you think your results would be so good in a larger area? Of course this is not at all a demerit of your work, not, so I think it is worth to be mentioned in the discussion.

RESPONSE: Thanks for the comment. Compared to similar studies that mapped tropical peatlands like Charters et al. (2019) and Crowson et al. (2019), we feel our approach is implemented on a relatively large and heterogeneous landscape. However, we agree that it could be tested on a larger area, and have therefore added a few sentences in the concluding part of the discussion, as recommended. There are also plans for end-users to implement the approach at the national level in Ghana.

Charters, L.J.; Aplin, P.; Marston, C.G.; Padfield, R.; Rengasamy, N.; Bin Dahalan, M.P.; Evers, S. Peat swamp forest conservation withstands pervasive land conversion to oil palm plantation in North Selangor, Malaysia. *Int. J. Remote Sens.* **2019**, *40*, 7409–7438, doi:10.1080/01431161.2019.1574996.
Crowson, M.; Warren-Thomas, E.; Hill, J.K.; Hariyadi, B.; Agus, F.; Saad, A.; Hamer, K.C.; Hodgson, J.A.; Kartika, W.D.; Lucey, J.; et al. A comparison of satellite remote sensing data fusion methods to map peat swamp forest loss in Sumatra, Indonesia. *Remote Sens. Ecol. Conserv.* **2019**, *5*, 247–258, doi:10.1002/rse2.102.

2. A more serious concern is that you don't seem to have a validation set. The test set should not be used to optimize the hyperparameters (ntree, mtry or feature sets) of a predictive model or it will not be useful to provide a proper error estimation as the model has been previously optimized using it. So, the most advisable option is to have 3 sets (training, validation and set). If you have optimized your model using the test set, I think you should call it validation test and produce a new test set to obtain the final accuracy measurement.

RESPONSE: Thanks for the comment, but we think there is some misunderstanding about the validation process. Data from polygon extents were used for model building (i.e. feature selection based on cross-validation and training) while the point-based data served as an independent sample for accuracy assessment. The text in Sections 2.3 and 2.5 has been modified to ensure clarity and consistency.